Purpose and Knowledge Base

The Laptop Advisor is a rule-based expert system that recommends laptop categories and specifications using forward chaining. It collects facts about budget, portability, battery life, gaming, creative work, office use, operating system preference, AI acceleration, screen size, and travel habits.

Implementation

In `engine.py`, initial facts are stored in a set called working memory. The `can_fire()` method checks that every condition of a rule is satisfied and its conclusion is not already known.

The `run()` method repeatedly selects an applicable rule, adds its conclusion to working memory, and records the reasoning step. Higher-priority rules fire first; equal priorities follow knowledge-base order. Newly derived facts can trigger additional rules. The process stops when no more rules can fire, and known conclusions are not derived again.

The `conclusions()` method separates recommendations, specifications, and other facts. The `main.py` file validates user input, runs the engine, and displays conclusions and an explanation of the fired rules.

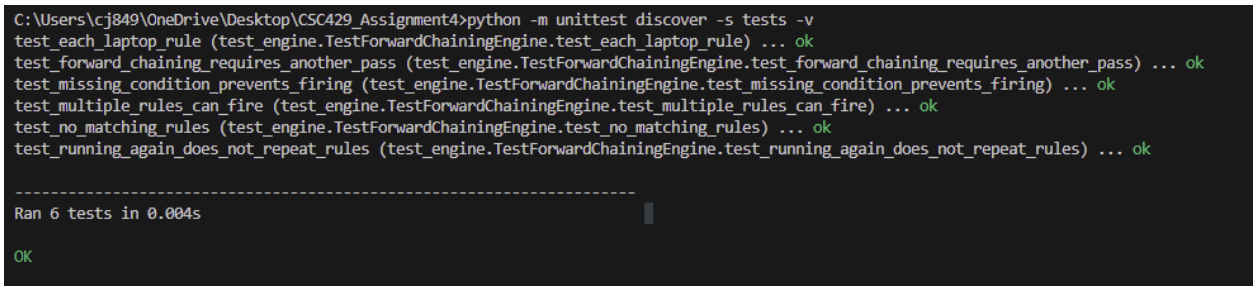
Testing and Results

All six automated test methods passed using:

python -m unittest discover -s tests -v

The tests checked all 10 laptop rules, missing conditions, multiple matching rules, unmatched inputs, repeated execution, and a two-rule chain in which a derived fact triggers another rule.

Figure 1. Automated test results



An interactive test was run using:

Python main.py

The answers specified a high budget, portability, long battery life, office-only use, frequent travel, and Windows. Gaming, creative work, AI acceleration, and a large screen were not selected.

Figure 2. Interactive run

```
C:\Users\cj849\OneDrive\Desktop\CSC429_Assignment4>python main.py
Laptop Advisor
Answer the questions to receive rule-based advice.

What is your budget? (low/medium/high): high
Is portability important? (y/n): y
Do you need long battery life? (y/n): y
Will you use the laptop for gaming? (y/n): n
Will you use it for creative work? (y/n): n
Is it only for office or basic productivity work? (y/n): y
Do you need AI acceleration? (y/n): n
Do you want a large screen? (y/n): n
Do you travel often? (y/n): y
Which operating system do you prefer? (windows/macos/linux): windows

Recommendations:
- premium_ultrabook

Suggested specifications:
- battery_60wh_plus

Explanation / fired rules:
- Premium Ultrabook: budget_high AND portable AND long_battery -> recommend:premium_ultrabook
- Travel + battery: travel_often AND long_battery -> spec:battery_60wh_plus
```

Limitations

The system recommends categories and specifications rather than specific laptop models. Some preference combinations may produce no recommendation, while others may produce multiple recommendations.

Rule priority controls firing order but does not rank the final recommendations or resolve conflicting preferences. The system uses definite facts and does not assign confidence scores.